Challenge Title: Automated RFP Intelligence & Commercial Proposal Generation System

Challenge Reference: CSTAM-OliveSoft

Company Collaborator: OliveSoft

Assigned Mentor:
- Khalil Masmoudi - OliveSoft AI & Data Engineer
- Bilel Djemel - OliveSoft AI & Data Engineer

1. Company Introduction & Challenge Context

Olivesoft operates in the IT Services & Software Engineering sector, specializing in enterprise digital solutions, web/mobile engineering, and custom software development. To drive digital transformation and address emerging business challenges, the company continuously explores innovations in Generative AI & LLMs, Multi-Agent Orchestration (n8n), and Retrieval-Augmented Generation (RAG).

In today's rapidly evolving software market, where rapid response times and tailored commercial proposals for public/private RFPs (Appels d'Offres) are increasingly critical, we currently rely on manual market monitoring, fragmented internal archives (past projects, client history, CV databases, tech stacks), and manually formatted commercial proposals. However, as our operations scale, we are actively seeking innovative approaches to automate tender discovery, conduct AI-driven prospect research, and generate customized commercial proposal presentations.

2. Problem Statement

Our main challenge is the time-intensive, manual process of identifying relevant RFPs, gathering prospect intelligence, aligning internal expertise (CVs, past projects, tools), and drafting comprehensive commercial proposals.
When a new tender or RFP is published or submitted, our teams experience delays in evaluation, high manual research overhead, and risk of suboptimal positioning due to fragmented access to internal knowledge assets. We require a robust solution capable of automatically detecting market leads, conducting agentic prospect research, retrieving matching Olivesoft capabilities via RAG, and generating customized commercial presentations and proposals without disrupting existing workflows.

3. Technical Mission

Challenge Objective:

To design, prototype, and demonstrate an end-to-end AI-powered system that monitors market tenders, enriches context through autonomous prospect research, leverages Olivesoft's internal knowledge base via RAG, and automatically generates tailored commercial proposal presentations using n8n orchestration and Generative AI.

Participating teams must build a system evaluated across the following core phases:

Phase 1: Core Functionalities MVP (Mandatory)

Total score: 50 points
Required Features:

- **Lead & Tender Detection Engine [15 Points]:** Automated ingestion and detection of published tenders/RFPs from public datasets, simulated feeds, or structured inputs.
- **Robust RAG Knowledge Retrieval [15 Points]:** A vector-search pipeline capable of indexing and extracting knowledge from Olivesoft's internal documentation (past project references, client portfolios, CV bank, tools, and technical stacks).
- **Agentic Prospect Research [10 Points]:** Autonomous AI agent workflow to research prospect details (sector, estimated revenue, past projects, key partners, specific domain requirements).
- **Workflow Orchestration with n8n [10 Points]:** Seamless orchestration using n8n to connect detection, research, RAG retrieval, and initial document generation.

Required Deliverables :

- System architecture diagram and data pipeline flow documentation
- Source code repository (including n8n workflows and RAG implementation)
- A short video demonstration of the MVP

Phase 2: Integration & Testing

Total score: 30 Points

Required Features:

- **Generative Presentation & Proposal Output [15 Points]:** Automated drafting of commercial proposal presentations matching Olivesoft's commercial templates and branding guidelines.
- RAG Benchmark & Retrieval Precision [10 Points]: Benchmarking and optimizing semantic search accuracy to ensure correct CV and project matching based on tender requirements.
- **Edge Case & Data Validation Handling [5 Points]:** Graceful handling of incomplete tender specifications, noisy web research data, or missing internal references.

Required Deliverables :

- RAG benchmark, evaluation report, and test suite execution results.
- Detailed integration & API workflow documentation.

Phase 3: UI/UX & Documentation

Total score: 10 Points

Refine the project for final evaluation by improving usability and preparing all required deliverables.

Required Features:

- **Quality of Interface / Usability [10 Points]:** Intuitive web interface/dashboard allowing sales managers to view detected leads, inspect matched internal assets, review prospect research, edit generated parameters, and download proposal decks.

Required Deliverables :

- Finalized UI/UX design and user experience dashboard.
- Technical report documenting project architecture, LLM/RAG strategies, and results.
- A demonstration video showcasing full end-to-end platform features and functionality.

Note:

- **Only the selected teams will advance to the Final Pitching phase**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**

Phase 4: Pitching

Total score: 10 Points

Present the completed solution to the company representatives and the evaluation committee during the CSTAM 3.0 event.

Required Deliverables :

- A professional presentation supporting the pitch.
- A live demonstration of the platform.
- Q&A session with the jury members.

Optional Advanced Features

Bonus Points

- **Multi-Format Deck Exporting [+2 Points]:** Direct generation and export to editable .pptx slides and formatted .pdf documents.
- **Automated Requirements Matrix & Score [+2 Points]:** Auto-generating a matrix scoring Olivesoft's technical coverage against RFP requirements.
- **Competitor & Budget Insights Agent [+2 Points]:** Agentic estimation of prospective budget ranges and competitive positioning analysis.
- **Human-in-the-Loop AI Refinement [+2 Points]:** An interactive UI feature allowing sales teams to prompt-edit specific slide sections in real-time.
- Multi-Source Crawler Integration [+2 Points]: Ingestion pipelines for real-time crawling across multiple public procurement portals.

4. Recommended Skills

You're encouraged to bring your own ideas, but familiarity with the following helps:

- **Hardware / Infrastructure:** Cloud
- Services, Docker Containers, Serverless Deployments.
- **Software & Frameworks:** n8n Workflow Automation, Python, LangChain / LlamaIndex, Fast API, React / Next.js / Tailwind CSS, Vector Databases (Qdrant, ChromaDB, Pinecone, or pgvector).
- **Protocols & Models:** REST APIs, Webhooks, OpenAI / Anthropic APIs, Open-Source LLMs (Ollama / vLLM), JSON-Schema formatting.

5. Rewards & Opportunities

- **Number of Winners:** 3 teams
- **Awards**
  - **First Place Team:** Summer internships and PFE 2027 opportunities at OliveSoft
  - **Second Place Team:** Certificate of Recognition
  - **Third Place Team:** Certificate of Recognition

Note:

- **Additional rewards may be announced later**
- **Teams will also compete for the Global CSTAM Book Grand Prizes Top 3 across all challenges**

General CSTAM Book Guidelines

While each team will tackle a unique industrial problem proposed by our collaborator partner companies, the core rules, expectations, deliverables, and support frameworks remain consistent across all projects.

1. Team Eligibility & Structure

Team Requirements:

- Open to all students.
- Eligible for IEEE members and Non IEEE Members.
- Each participant may join **only one team.**
- Each team may apply for **only one technical challenge.**

Team Composition:

- Teams must consist of **3 to 5 members.**
- **Bonus Points:** Teams composed of members from **different levels of study** will receive a **bonus of 5 points** during the evaluation process.
- Interdisciplinary teams are strongly encouraged.

2. CSTAM Book Timeline

2.1. Initial Registration:

Registration Opens: August 20 , 2026

Teams must apply through the official CSTAM Book Registration Form, where they are required to :

- Provide the team's general information.
- Rank their preferred technical challenges.
- Provide their related information to preferred project
- Accept and sign the **CSTAM Book Participant Commitment,**acknowledging the rules, responsibilities, and terms governing the CSTAM Book.

2.2. Infosessions:

Prior to the project selection phase, each partner company will host an information session to present its technical challenge and answer participants' questions, helping them make informed decisions when ranking their project preferences.

2.3. Challenge Selection Phase:

Acceptance into a specific company challenge will be evaluated based on:

- Participant CVs and academic backgrounds.
- A brief statement of motivation.
- Evidence of previous similar projects or relevant technical experience.

Note: Accepted teams will be notified via email.

2.4. Projects Development Phase (Phase 1 & 2):

Launching Date: Upon announcement of the selected teams

The challenge follows a strictly phased workflow to ensure steady progress. Throughout the project development phase, teams must actively develop their solutions, meet the objectives and deliverables of each phase, and adhere to the established timeline.

The following schedule defines the key phases and deadlines:

Phase 1 : Core Functionalities MVP (Mandatory)

Total score: 50 points
Deadline: October 1, 2026

Teams are expected to deliver the initial version of their solution, including the core functionalities and proposed technical approach.

Phase 2 : Integration & Testing

Total score: 30 points
Deadline: October 20, 2026

Teams will focus on integrating project components, improving performance, and validating the functionality of their solution.

Phase 3 : UI/UX & Documentation (Selection Phase)

Total score: 10 points
Deadline: November 1, 2026

Only selected teams will advance to the Pitching phase and selection criterias will be based on:

- Source code repository
- Proof of Concept (PoC)
- **Technical Report Requirements:**
  - Abstract
  - Team Overview & Contributions
  - Problem Statement & Industrial Context
  - Proposed Solution
  - System design
  - Mockups
  - Technologies, Frameworks, and Tools Used
  - Development Milestones
  - Testing results
  - Future Improvements
  - References (if applicable)
- Commitment to the Phases required deliverables
- Engagement and commitment in mentoring sessions
- **Late Submission Penalty:** 5 Points will be deducted for the late submission of required delivera-

Note: Accepted teams will be notified via email.

Phase 4 : Pitching

Total score: 10 points
Deadline: In Person at CSTAM 3.0

Selected teams will present their solutions before the company representatives and the evaluation committee through a final pitch and live demonstration during the CSTAM 3.0 event.

The following guidelines apply to the pitching sessions:

- **Language Requirement:** All presentation, Technical Report, demonstration, and Q&A sessions must be conducted strictly in **English.**
- **Team Representation:** A maximum of **two (2)** team members may participate in the final pitching phase.
- **Format & Time Allocation for the Presentation (12 Minutes in Total):**
    - **5 minutes:** Presentation
    - **2 minutes:** Live Demonstration
    - **5 minutes:** Q&A Session

Optional Bonus

- **Total score: Up to 12 points**

Rewards exceptional creativity, "outside-the-box" engineering, or advanced stretch goals. Judges may award bonus points for complex, secondary features that exceed basic MVP requirements to significantly elevate the project.

Winners Announcement

- **In Person at CSTAM 3.0**
- **Number of Winners:** 3 teams
- **Awards**
    - **First Place Team:** Summer internships and PFE 2027 opportunities at Company Challenge Owner
    - **Second Place Team:** Certificate of Recognition
    - **Third Place Team:** Certificate of Recognition

Note:

- Additional rewards may be announced later
- Teams will also compete for the Global CSTAM Book Awards Top 3 across all subjects

3. Global CSTAM Book Awards :

The scores from all company challenges will be centralized to establish the overall CSTAM Book ranking.

The highest scoring teams will be awarded the 1st, 2nd, and 3rd Place Overall CSTAM Book Titles based on the following criteria:

- **Overall Project Score:** Final score obtained from all challenge phases.
- **Technical Excellence:** Quality of implementation, architecture, and technical solution.
- **Innovation & Impact:** Level of originality and added value.
- **Project Completion & Deliverables Quality:** Compliance with requirements, documentation quality, and maturity of the final solution.
- **Presentation & Demonstration Performance:** Quality of the final pitch, clarity of communication, and effectiveness of the live demonstration.
- **Team Commitment & Collaboration:** Engagement, progress throughout the challenge, and ability to work effectively as a team.

4. AI Usage Policy :

The use of Artificial Intelligence (AI) tools is permitted throughout the development of the project, provided that the following guidelines are respected:

- Teams may use AI-assisted tools (e.g., ChatGPT, GitHub Copilot, Claude, Gemini) to support research, development, debugging, documentation, and design activities.
- All AI-generated code, content, or design artifacts **must be carefully reviewed, validated, and understood by the team before submission.**
- Teams are fully responsible for the accuracy, originality, security, and quality of all submitted deliverables, regardless of the tools used during development.
- During the final presentation and Q&A session, participants must be able to clearly explain and justify any AI-assisted contributions.

Note: The use of AI tools does not replace the team's technical understanding or expertise.

5. Mentoring Sessions & Technical Support

Throughout the project development journey, each team will benefit from guidance provided by a dedicated technical mentor from the corresponding company partner.

Mentoring sessions will combine **workshop sessions and project supervision.**

Role of Mentors:

The mentor will support teams by:

- Providing clarification on project objectives, requirements, and expected deliverables.
- Guiding teams in defining appropriate technical approaches and solution architectures.
- Following project progress, providing feedback, and helping resolve technical challenges.
- Supporting teams during implementation, integration, testing, and validation stages.
- Sharing technical insights and best practices to ensure alignment with phases requirements.

Role of Teams:

- Mentoring sessions are **mandatory for all teams** throughout the project development phases.
- Each team must ensure that its **Team Leader** represents the team during the supervision sessions.
- The Team Leader will be responsible for sharing the team's progress, challenges and questions with the mentor during the supervision session.
- Mentoring session attendance, participation, and engagement will be taken into consideration as part of the final evaluation criteria

Note: Teams will be notified via email about their mentoring sessions.

6. Participant Commitment

By registering and proceeding with a CSTAM Book challenge, all team members must agree to the **CSTAM Book Participant Commitment Agreement.**